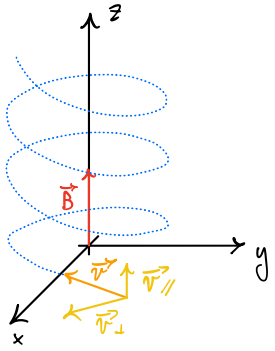
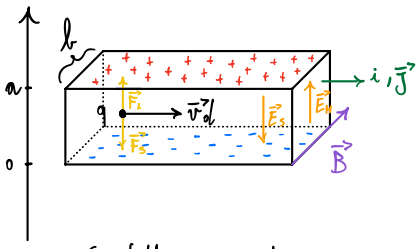
Osserviamo:




## EFFETTO HALL 1879




Conduttore simmetrico

$$\vec{F}_L = q\,\vec{v}_d \times \vec{B} = q\,v_d\,B\,\vec{u}_z$$

$$\vec{F}_S = q\,\vec{E}_S = q\,E_S\,(-\vec{u}_z)$$

$$\Big\} \quad \vec{F}_S + \vec{F}_L = 0$$

oss. È come se ci fosse una f.e.m. che separa le cariche

$$\vec{E}_H = -\vec{E}_S = v_d\,B\,\vec{u}_z$$

$$\Rightarrow \text{f.e.m.} = \int_0^a \vec{E}_H \cdot d\vec{z} = v_d\,B\,a$$

$$= -\int_0^a \vec{E}_S \cdot d\vec{z} = \int_0^a dV = V_a - V_0 = V^+ - V^- = \Delta V_H$$

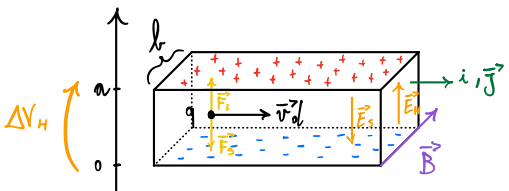
$$\Big\} \quad \Delta V_H = v_d\,B\,a$$

$$\Delta V_H = v_d\,B\,a = \frac{J}{nq}\,B\,a = \frac{i}{nqS}\,B\,a = \frac{i}{nqab}\,B\,a = \frac{i\,B}{nq\,b} \equiv \frac{R_H\,i\,B}{b} \Rightarrow \boxed{\Delta V_H = R_H\,\frac{i\,B}{b}}$$

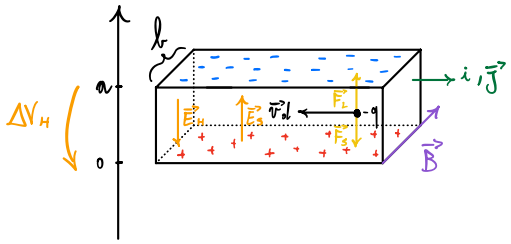
Potenziale di Hall

$$\boxed{R_H = \frac{1}{nq}}$$

Costante di Hall



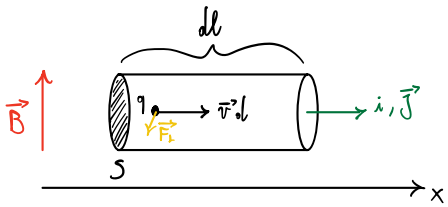
• Se fosse la carica negativa a spostarsi:




Sperimentalmente si è ottenuto $\downarrow \Delta V_H$, dunque possiamo affermare che sono le cariche negative a muoversi

• $\Delta V_H = R_H \frac{iB}{b} \Rightarrow B = \frac{\Delta V_H b}{R_H \cdot i}$ (Sonda di Hall, mi permette di trovare B)

• Consideriamo:




$d\tau = S\, dl$

$d\vec{F} = n\, d\tau\, q\, \vec{v}_d \times \vec{B} = n S\, dl\, q\, v_d\, \vec{u}_x \times \vec{B}$

$= n q v_d S\, dl\, \vec{u}_x \times \vec{B} = J S\, dl\, \vec{u}_x \times \vec{B} = i\, d\vec{l} \times \vec{B}$

$\Rightarrow \boxed{d\vec{F} = i\, d\vec{l} \times \vec{B}}$ II equazione elementare di Laplace

$\vec{F}_L = q\, \vec{v} \times \vec{B}$

$\vec{F} = \int i\, d\vec{l} \times \vec{B}$

} sono equivalenti? Sì!

$q\vec{v} = i\, d\vec{l}$

$q v = i\, dl$

$n q\, d\tau\, v = i\, dl$

$J\, d\tau = i\, dl$

$\frac{i}{S} d\tau = i\, dl$

$\frac{i}{S} \cdot S\, dl = i\, dl$

$\boxed{i\, dl = i\, dl}$

$\boxed{\vec{F} = \int_l i\, d\vec{l} \times \vec{B}}$ Forza di Laplace $\Rightarrow$ per un circuito $\vec{F} = \oint i\, d\vec{l} \times \vec{B}$

• Per $\vec{B}$ uniforme e $\gamma$ rettilineo

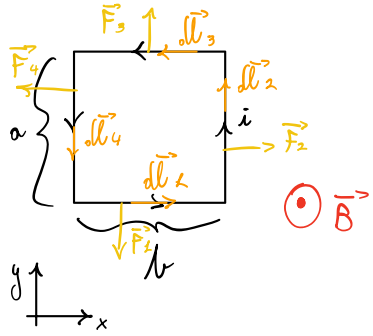
$\vec{F} = \int_l i\, d\vec{l} \times \vec{B} = \left( \int_\gamma i\, d\vec{l} \right) \times \vec{B} = i \vec{l} \times \vec{B}$ $\quad [T] = \left[ \frac{N}{A \cdot m} \right]$

· $\vec{B}$ uniforme e $\gamma$ chiuso

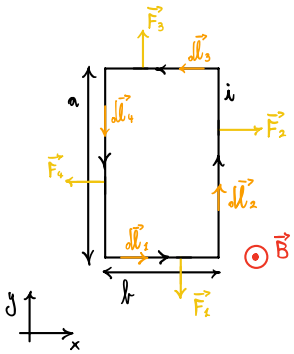
$$\vec{F} = \oint i\, d\vec{l} \times \vec{B} = i\left(\oint d\vec{l}\right) \times \vec{B} = 0$$

($\oint d\vec{l} = 0$)

$\Rightarrow \vec{F}_1$ è nulla su un circuito chiuso!




– Esempio



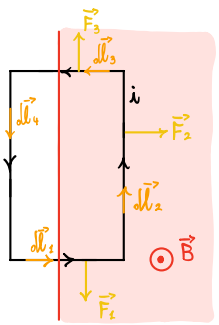

$$\vec{F}_1 = i\, b\, B\, (-\vec{u}_y)$$

$$\vec{F}_2 = i\, a\, B\, \vec{u}_x$$

. . . .

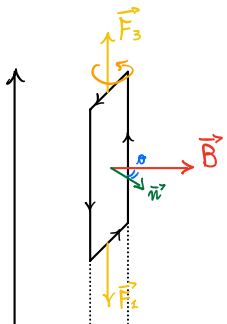
$\Rightarrow \vec{F}_1 = -\vec{F}_3$ e $\vec{F}_2 = -\vec{F}_4$

Se la spira fosse solo per metà immersa nel campo:




$\vec{F}_2$ spingerebbe tutta la spira nel campo, fino ad una situazione di equilibrio

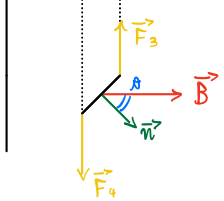
· Consideriamo:




$$\vec{F}_1 = -\vec{F}_3 \quad \text{e} \quad |\vec{F}_2| = |\vec{F}_4| = i\,B\,a$$

$$\vec{M} = i\,B\,a\,b\,\sin\theta\,\vec{u}_z = i\,a\,b\,B\sin\theta\,\vec{u}_z = i\,S\,\vec{n}\times\vec{B}$$

($\vec{n}$)



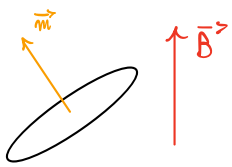

$\Rightarrow$ $\boxed{\vec{M} = \vec{m} \times \vec{B}}$

055. Con il dipolo avevamo visto $\vec{M} = \vec{p} \times \vec{E}$

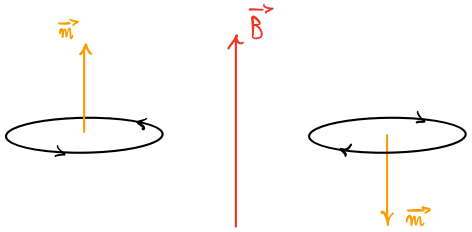
$\Rightarrow$ $\boxed{\vec{m} = i S \vec{n}}$ Momento di dipolo magnetico

$\vec{n}$ determinato con la regola della mano destra (verso di percorrenza della corrente)

• Considero:




$\Rightarrow$

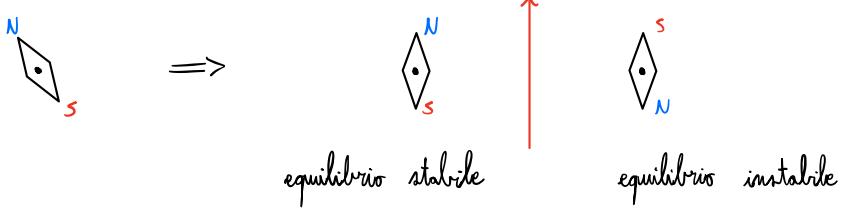



Il campo tende ad allineare la spira percorsa da carica

equilibrio stabile equilibrio instabile

$\begin{cases} \vec{M} = \vec{p} \times \vec{E} \Rightarrow U = -\vec{p} \cdot \vec{E} \\ \vec{M} = \vec{m} \times B \Rightarrow U = -\vec{m} \cdot \vec{B} \end{cases}$

Lo stesso accade per un ago:

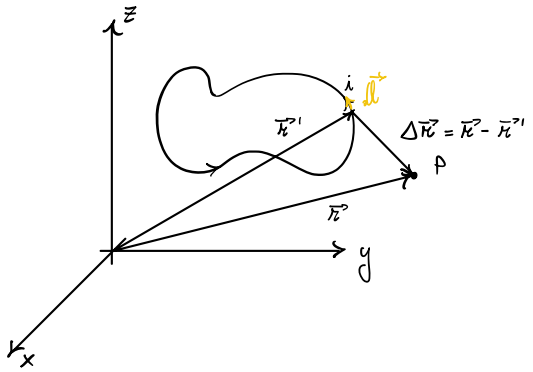



equilibrio stabile equilibrio instabile

$$\cdot \vec{F} = -\vec{\nabla} U = -\vec{\nabla}\left(-\vec{m}\times\vec{B}\right) = \vec{\nabla}\left(\vec{m}\times\vec{B}\right) = \left(\vec{m}\cdot\vec{\nabla}\right)\vec{B}$$

$$m_x \cdot \frac{\partial}{\partial x} + m_y \cdot \frac{\partial}{\partial y} + m_z$$

## CAMPI GENERATI DA CIRCUITI NEL VUOTO



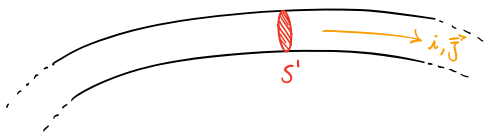

$$d\vec{B} = \frac{\mu_0}{4\pi}\, i\, \frac{d\vec{l}\times\Delta\vec{r}}{|\Delta\vec{r}|^3}$$

I Legge di Laplace (Biot - Savart)

$$\vec{B}_0 = \frac{\mu_0}{4\pi}\, i \oint_{\mathcal{C}} \frac{d\vec{l}\times\Delta\vec{r}}{|\Delta\vec{r}|^3}$$

Campo generato da un circuito

Consideriamo




$$i = \Phi_{S'}(\vec{J}) = \int_{S'} \vec{J}(\vec{r}\,')\cdot\vec{n}\, dS'$$

$$\Rightarrow B_0 = \frac{\mu_0}{4\pi}\oint_{\mathcal{C}}\left(\int_{S'}\vec{J}(\vec{r}\,')\cdot\vec{n}\, dS'\right)\frac{d\vec{l}'\times\Delta\vec{r}}{|\Delta\vec{r}|^3} = \frac{\mu_0}{4\pi}\int_{\mathcal{T}}\frac{\vec{J}(\vec{r}\,')\times\Delta\vec{r}}{|\Delta\vec{r}|^3}\, d\tau$$

$\mu_0 = 4\pi\cdot 10^{-7}$ $\left[\frac{N}{A^2}\right] = \left[\frac{H}{m}\right]$ $H$ = Henry $(= \Omega\cdot s)$ $\Rightarrow$ Permeabilità magnetica del vuoto

$$C\ \frac{V}{m}\cdot\frac{L}{A\cdot\frac{C}{S}} = \frac{VS}{Am} = \frac{\Omega s}{m}$$